



New Zealand Diploma in Information Technology Technical Support Level 5

DDM512: Database Design and Management

Assessment: DDM512 Part 1 v1

Project



Learner Name	Deacon George
Learner ID	AEU3001969
Assessor Name	Tracy Fei
Submission Result	

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Task 1 – Project Planning & Analysis

Task/Mahi 1.1 – Business Overview

1.1.1 Business Overview

1. Delivery of Tertiary Education.

Wawatahia University's primary service is the delivery of higher education. This is accomplished through a wide range of academic courses. This service includes

- Design and delivery of course content
- Providing lectures, tutorials, labs, and assessments
- Supporting students through academic guidance
- Ensuring each course meets National Education and Quality standards

This University exists to educate students and help them on their academic journey towards qualification. With over 200 students enrolled, this service is central to the University's mission and daily operations. Every course also requires a project. Teaching and project-based learning are tightly integrated into the academic experience.

2. Project Supervision and Academic Research Support

A second major service is the supervision and assessment of student projects, which every student must complete as part of their course requirements.

Faculty Professors:

- Guide students through project planning and execution
- Provide academic supervision and feedback
- Assess project outcomes
- Ensure projects meet academic integrity and research standards

Every course requires a project, and each student works on one project that is assessed by a professor. This makes project supervision a major operational service.

1.1.2. Two Main Mission Statements

Mission Statement 1: Enhance Accuracy and Accessibility of Student and Project Information

The mission of the Wawatahia University's Database System is to provide a secure, centralised, and highly accurate platform for storing and managing all student, course, professor, and project information. The system aims to ensure that authorised staff can quickly access up to date records, enabling efficient academic supervision, improved decision-making, and compliance with New Zealand data protection standards.

Mission Statement 2: Streamline Academic Workflows and Strengthen Project Supervision Processes

The mission of the database system is to streamline the university's operational processes by automating the management of student enrolment's, course allocations, and project supervision activities. The system will support professors and administrative staff by reducing manual workload, eliminating duplication, and enabling fast retrieval of project details, ultimately improving the quality and consistency of academic support to students.

1.1.3 Two Benefits

1. Improved Efficiency and Accuracy in Managing University Records.

The new database system will significantly streamline Wawatahia University's daily academic and administrative operations by centralising all relevant information into a single, structured platform. This will reduce manual data entry, eliminate duplicated records, and will ensure that staff can quickly retrieve accurate information when enrolling students, assigning professors, or tracking project progress.

A centralised and well-designed database helps the University comply with the New Zealand Privacy Act 2020. This requires organisations to store personal information securely, ensure data accuracy, and restrict access to authorised personnel only.

2. Enhanced Data Security, Integrity, and System Reliability

The new database system improves the University's information system by providing modern management capabilities, including relational integrity, secure authentication, automated backups, and structured relationships between students, professors, courses, projects, and faculties. This results in a more reliable, scalable, and maintainable system that supports long-term growth.

The improved database architecture strengthens compliance with the Privacy Act 2020 and New Zealand Educational Data Governance Standards. This is accomplished by:

- Enforcing referential integrity which ensures student and project records cannot be lost, mismatched, or corrupted.

- Supporting secure storage of sensitive personal information through modern database security features.
- Allowing regular backups reduces the risk of data loss, ensuring business continuity.
- Providing role-based access controls, ensuring only authorised staff can view or modify sensitive information.
- Ensuring data minimisation and accuracy as required under the Privacy Act.

By improving the reliability and security of the information system itself, the University reduces compliance risks and ensures that its data management practices meet modern legal and technical standards.

Task/Mahi 1.2 – Business Requirements

1.2.1 Three Functional Requirements

Data Management & Manipulation:

- The database will allow users to add, update, delete, and retrieve data. This includes support for various data types (text, numeric, date/time) and structuring the schema to store different records.

Querying and Searching:

- The system supports efficient searching and sorting functionality, enabling users to find data based on specific criteria.

Reporting and Exporting:

- Generating reports and exporting data in various formats (e.g., CSV, PDF).

1.2.2 One Non-Functional Requirement for Usability & One for Security

Performance:

- Speed and responsiveness define how fast the database returns query results. Examples include transaction throughput (e.g., 10,000 transactions/second) and latency (e.g., response time under 1 second). Long response times can result in a negative user experience and inefficiencies.

Security and Data Integrity:

- Protects data against unauthorised access using encryption, user login, role-based access control, and compliance with regulations.

Task/Mahi 1.3 – Technical and Administrative Details

1.3.1 Three Database Management Systems Researched

MySQL (Relational DBMS) is a structured, table-based relational database that uses SQL for querying. It's widely used to manage organised data such as enrolments, course records, and assessment information. Its strong SQL support and reliability make it suitable for systems that require consistent, accurate data storage.

MongoDB (Document-oriented NoSQL DBMS) stores information as flexible JSON-style documents, which allows a university to manage unstructured or rapidly changing data. For example, student activity logs, research datasets, or dynamic web content. It's schema-less design supports projects where data formats evolve over time.

Neo4j (Graph DBMS) represents data as nodes and relationships, making it ideal for modelling complex university networks such as academic collaborations, research citations, or student-course-faculty connections. It excels when the focus is on analysing relationships rather than strong storing traditional tabular data.

1.3.1.1 Recommendation to the Organisation & Justification

MySQL is the most suitable DBMS for this University project as it provides a stable, structured, and academically recognised platform for managing organised data such as student records, course information, enrolments, and assessment results. Its relational model ensures high data integrity, which is essential in an educational setting where accuracy and consistency are non-negotiable.

From an operational perspective, MySQL is widely supported across university IT infrastructures, integrates easily with learning management systems, and offers strong security features for protecting sensitive student data. It is also open-source, making it cost effective for both teaching and institutional deployment.

Overall, MySQL offers the right balance of reliability, performance, and academic suitability, making it the recommended choice for a university-level database system.

1.3.2 Two Optimisation Techniques & Justification

Defragmenting Tables:

- Over time, as data is added, removed, or updated within database tables, fragmentation can occur. Regularly defragmenting tables helps in reclaiming unused space and improving query performance. In MySQL, this can be done using the OPTIMIZE TABLE command, which essentially rebuilds the table to eliminate fragmentation

Archiving Old Data:

- In databases where data accumulates over time, it's beneficial to archive old data that is no longer actively used. Archiving helps in reducing the size of the active database, which can lead to faster backups, quicker maintenance tasks. And improved overall performance. Archived data can be stored in a separate data base or a different storage medium for historical analysis or record-keeping purposes.

1.3.3 Four Storage Options Evaluated

Hard Drives / Solid State Drives:

- HDDs and SSDs store data permanently. Universities typically use this layer for hosting database files, system backups, and application data because it offers large capacity at a reasonable price.

Cloud Storage:

- Cloud storage services (e.g., AWS S3, Azure Blob Storage, Google Cloud Storage) provide scalable, remote data storage. Universities use cloud storage for research datasets, online learning platforms, and distributed systems due to its flexibility, redundancy, and global accessibility.

Network Attached Storage (NAS):

- NAS is a dedicated storage device connected to the University network. It allows multiple users and systems to access shared files. It is commonly used for departmental data, shared teaching resources, and collaborative research projects.

Magnetic Tape:

- Tape storage is a long-term, high-capacity archival storage option that universities still use today. Viable as a cost-effective-per-terabyte option. It has a very long lifespan of 20-30 years, and each cartridge can store 12-30 terabytes of data. Not suitable for daily operational databases or data retrieval so is strictly archival.

1.3.3.1 Two Storage Options for Backing Up Database & Justification

1. Solid State Drives (SSD)

SSDs use flash-based storage rather than spinning disks found in HDDs, allowing for significantly faster read/write speeds. In a university environment, SSDs are ideal for hosting active databases, virtual machines, and learning management systems because they reduce query response times and improve overall system performance. Their durability and low failure rates also make them suitable for critical academic systems that require high availability.

Justification:

SSDs are recommended when the university needs high-speed access to frequently used data such as student information, course enrolment, or research applications that process large datasets. Their speed directly improves user experience for students and staff, while their reliability reduces downtime and maintenance overhead.

2. Cloud Storage

Cloud storage provides scalable, remote, and redundant data storage through platforms such as AWS, Azure, or Google Cloud. Universities use cloud storage to support online learning systems, research collaboration, and long-term data retention. Cloud platforms offer built-in redundancy, automated backups, and global accessibility, making them suitable for distributed academic environments.

Justification

Cloud storage is recommended when the university requires flexibility, scalability, and cost-effective expansion. It is particularly valuable for storing large research datasets, hosting online course materials, and supporting hybrid learning environments. Cloud services also reduce the need for on-premises hardware. This lowers maintenance costs and improves disaster recovery capabilities.

1.3.4 Data Backup Plan**PURPOSE**

The purpose of this plan is to ensure the universities critical data is protected from loss caused by hardware failure, cyberattacks, human error, or system corruption. A structured daily backup routine supports business continuity and ensures rapid recovery when required.

BACKUP TECHNIQUES**1. DIFFERENTIAL BACKUP**

A differential backup copies all data that has changed since the last full backup. The file grows larger each day but is faster to restore than incremental. It is ideal for capturing daily changes in a predictable way.

2. INCREMENTAL BACKUP

An incremental backup stores only the data that has changed since the previous backup. This reduces storage usage and speeds up the backup process. This method is widely recommended for daily operations because it is efficient and supports fast, automated scheduling.

DAILY BACKUP SCHEDULE

Backup Type	Frequency	Time	Storage Location
Incremental Backup	Daily	2:00 AM	Cloud
Differential Backup	Daily	11.00 PM	SSD - Cloud

STORAGE OPTIONS

1. Local SSD Storage

- Stores the daily differential backup.
- SSDs provide high-speed access and low latency, making them suitable for operational restoration.
- Supports immediate recovery in case of minor data loss events.

2. Cloud Storage

- Geographically distributed redundancy
- Automated scheduling, replication, and secure remote access.
- Protection against disasters, theft, or hardware failure.

RETENTION POLICY

- Daily incremental backups: Retain for 14 days
- Daily differential backups: Retain for 7 days

BACKUP VERIFICATION

- Perform weekly restore tests to confirm data can be recovered.
- Use automated integrity checks (checksums) after each backup.
- Review backup logs daily for errors or failures.

1.3.5 Three Professional & Ethical Practices

Access Control and Authentication.

A core responsibility is ensuring that only authorised staff can access sensitive organisational data.

- This can be achieved by implementing Role-Based Access Control (RBAC) so users can only access data required for their job.
- The use of Multi-Factor Authentication (MFA) for all database logins is a good, layered approach to secure login.
- Use unique user accounts rather than shared credentials to maintain accountability.
- Regularly review and update access permissions, especially when staff change roles or leave.

Ethical and Legal Compliance

- Prevents unauthorised access to personal and confidential information.
- Supports compliance with privacy regulations such as the New Zealand Privacy Act 2020, which requires organisations to protect personal data from misuse or disclosure.
- Ensures transparency and accountability in how data is accessed and used.

Apply Data Encryption and Secure Storage Practices

Protecting data both in transit and at rest is essential for maintaining confidentiality and integrity.

- Encrypt sensitive data stored in the database using strong encryption standards such as AES-256.
- Use TLS/SSL to encrypt data transmitted between applications and the database server.
- Store backups in secure locations (cloud).
- Implement regular patching and updates to database software to remove vulnerabilities.

Ethical and Legal Compliance

- Ensures personal and organisational data cannot be intercepted or read by unauthorised parties.
- Meets legal obligations to safeguard sensitive information and reduce risks of data breaches as brought out in Principle 5 of the Privacy Act.

Maintain Audit Logging, Monitoring, and Compliance Documentation

Ethical database management requires transparency, traceability, and adherence to organisational and legal standards.

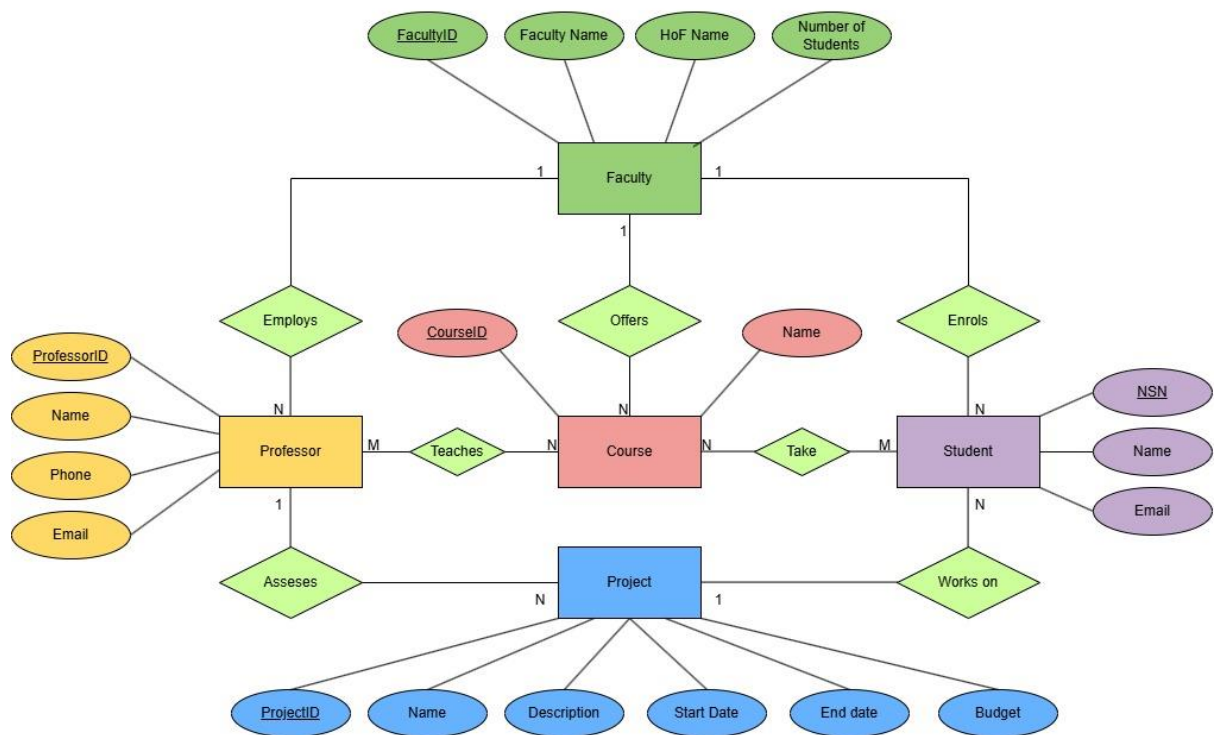
- Enable audit logs to track user activity including access attempts.
- Monitor logs regularly for suspicious behaviour or potential security incidents.
- Document all database procedures, backup routines, and security controls.
- Conduct periodic compliance checks and internal audits to ensure policies are being followed.

Ethical and Legal Compliance

- Provides evidence of responsible data handling and supports investigations if issues arise.
- Helps the organisation to meet regulatory requirements for accountability and reporting.
- Ensures the database is managed in a way that is transparent, ethical, and aligned with organisational policies.

Task 2 – Database Design

Task/Mahi 2.1 – ER Diagram



Task/Mahi 2.2 – Normalisation

2.2.1 Process of Normalising Data (UNF, 1NF – 3NF)

1NF:

1NF	FacultyID	Faculty Name	Hof Name	Number of Students	NSN	Name	Email	CourseID	Name	ProfessorID	Name	Phone	Email	ProjectID	Name	Description	Start Date	End Date	Budget
	NZU712	Advanced Automotive	Ken Block	722	NSN051155	Martin McFly	martym@refusion.com	AE606	Automotive Engineering Lvl 6	WU1955	Emmett Brown	07-544-5042	E.Brown@WU.ac.nz	HCT12	Hydrogen Cell technology	Design and build a next generation hydrogen cell engine	25/05/2026	03/08/2026	\$16,500.00
	NZU712	Advanced Automotive	Ken Block	722	NSN051155	Martin McFly	martym@refusion.com	TM888	Temporal Mechanics 101	WU1955	Emmett Brown	07-544-5042	E.Brown@WU.ac.nz	HCT12	Hydrogen Cell technology	Design and build a next generation hydrogen cell engine	25/05/2026	03/08/2026	\$16,500.00
	NZU404	Theoretical Physics	Sheldon Cooper	518	NSN897654	Rangi Mason	ranges12@bills.co.nz	APL67	Gravity and Physics	WU1643	Isaac Newton	07-566-2131	I.Newton@WU.ac.nz	TBT88	Tractor Beam Technology	Apply graviton and physics knowledge to design, build, and test a tractor beam	19/09/2026	25/12/2026	\$22,500.00
	NZU404	Theoretical Physics	Sheldon Cooper	518	NSN211015	Aria Thompson	anton@graffy.com	ME404	Mechanical Engineering Lvl 4	WU1905	Albert Einstein	07-574-8082	A.Einstein@WU.ac.nz	CRO17	Chronon Cultivation and Tachyon Harvesting	Design and build an apparatus to cultivate chronon particles and harvest tachyon particles to be used in temporal tunneling	01/06/2026	30/09/2026	\$37,000.00
	NZU404	Theoretical Physics	Sheldon Cooper	855	NSN24996	Liam Faulkner	liamian@fipso.co.nz	ME404	Mechanical Engineering Lvl 4	WU1905	Albert Einstein	07-574-8082	A.Einstein@WU.ac.nz	CRO17	Chronon Cultivation and Tachyon Harvesting	Design and build an apparatus to cultivate chronon particles and harvest tachyon particles to be used in temporal tunneling	01/06/2026	30/09/2026	\$37,000.00
	NZU767	Warp Theory	Leah Brahm	855	NSN555667	Hana Patel	hmutothere@geignone.com	WT808	Warp Theory Tier 8	WU1701	Geordie LaForge	07-448-8555	G.LaForge@WU.ac.nz	S67	Structural Integrity	Design a structural integrity field to be used in starship integrated systems	04/05/2026	27/07/2026	\$45,000.00
	NZU767	Warp Theory	Leah Brahm	855	NSN007007	Hemi Bond	hmapzy@gotch.com	WT808	Warp Theory Tier 8	WU1701	Geordie LaForge	07-448-8555	G.LaForge@WU.ac.nz	WPC46	Warp Plasma Containment	Design magnetic containment for warp plasma	01/05/2026	26/07/2026	\$12,500.00

2NF:

2NF

Faculty Table				Professor Table			
FacultyID	Faculty Name	Hof Name	Number of Students	ProfessorID	Name	Phone	Email
NZU712	Advanced Automotive	Ken Block	722	WU1955	Emmett Brown	07-544-5042	E.Brown@WU.ac.nz
NZU404	Theoretical Physics	Sheldon Cooper	518	WU1905	Albert Einstein	07-574-8082	A.Einstein@WU.ac.nz
NZU767	Warp Theory	Leah Brahm	855	WU1643	Isaac Newton	07-566-2131	I.Newton@WU.ac.nz
				WU1701	Geordie LaForge	07-448-8555	G.LaForge@WU.ac.nz

Project Table				Course			
ProjectID	Name	Description	Start Date	End Date	Budget	ProfessorID(FK)	CourseID
HCT12	Hydrogen Cell technology	Design and build a next generation hydrogen cell engine	25/05/2026	03/08/2026	\$16,500.00	WU1955	AE606
TBT88	Tractor Beam Technology	Apply graviton and physics knowledge to design, build, and test a tractor beam	19/09/2026	25/12/2026	\$22,500.00	WU1643	ME404
CRO17	Chronon Cultivation and Tachyon Harvesting	Design and build an apparatus to cultivate chronon particles and harvest tachyon particles to be used in temporal tunneling	01/06/2026	30/09/2026	\$37,000.00	WU1905	TM888
S67	Structural Integrity	Design a structural integrity field to be used in starship integrated systems	04/05/2026	27/07/2026	\$45,000.00	WU1701	WT808
WPC46	Warp Plasma Containment	Design magnetic containment for warp plasma	01/05/2026	26/07/2026	\$12,500.00	WU1955	APL67

Student Table			
NSN	Name	Email	FacultyID(FK)
NSN051155	Martin McFly	martym@refusion.com	NZU712
NSN211015	Aria Thompson	anton@graffy.com	NZU404
NSN897654	Rangi Mason	ranges12@bills.co.nz	NZU404
NSN555667	Hana Patel	hmutothere@geignone.com	NZU767
NSN24996	Liam Faulkner	liamian@fipso.co.nz	NZU404
NSN007007	Hemi Bond	hmapzy@gotcha.com	NZU767

Student_Course		Professor_Course	
NSN(FK)	CourseID(FK)	ProfessorID(FK)	CourseID(FK)
NSN051155	AE606	WU1955	AE606
NSN051155	TM888	WU1955	TM888
NSN897654	APL67	WU1643	APL67
NSN211015	ME404	WU1905	ME404
NSN24996	ME404	WU1701	WT808
NSN555667	WT808		
NSN007007	WT808		

3NF:

3NF														
Faculty Table					Professor Table									
FacultyID	Faculty Name	Hof Name	Number of Students		ProfessorID	Name	Phone	Email	FacultyID(FK)					
NZU712	Advanced Automotive	Ken Block	722		WU1955	Emmett Brown	07-544-5042	E.Brown@WU.ac.nz	NZU712					
NZU404	Theoretical Physics	Sheldon Cooper	518		WU1905	Albert Einstein	07-574-8082	A.Einstein@WU.ac.nz	NZU404					
NZU767	Warp Theory	Leah Brahm	855		WU1643	Isaac Newton	07-566-2131	I.Newton@WU.ac.nz	NZU404					
					WU1701	Geordie LaForge	07-448-8555	G.LaForge@WU.ac.nz	NZU767					
Project Table					Course					Student Table				
ProjectID	Name	Description	Start Date	End Date	Budget	ProfessorID(FK)	CourseID	Name	FacultyID(FK)	SN	Name	Email	FacultyID(FK)	ProjectID(FK)
HCT12	Hydrogen Cell Technology	Design and build a next generation hydrogen cell engine	25/05/2026	03/08/2026	\$16,500.00	WU1955	AE606	Automotive Engineering Lvl 6	NZU712	NSN051155	Martin McFly	martym@refusion.com	NZU712	HCT12
TBT88	Tractor Beam Technology	Apply graviton and physics knowledge to design, build, and test a tractor beam	19/09/2026	25/12/2026	\$22,500.00	WU1643	ME404	Mechanical Engineering Lvl 4	NZU404	NSN211015	Aria Thompson	anton@graffy.com	NZU404	CRO17
CRO17	Chronon Cultivation and Tachyon Harvesting	Design and build an apparatus to cultivate chronon particles and harvest tachyon particles to be used in temporal tunneling	01/06/2026	30/09/2026	\$37,000.00	WU1905	TM888	Temporal Mechanics 101	NZU712	NSN897654	Rangi Mason	ranges12@bills.co.nz	NZU404	TBT88
S67	Structural Integrity	Design a structural integrity field to be used in starship integrated systems	04/05/2026	27/07/2026	\$45,000.00	WU1701	WT808	Warp Theory Tier 8	NZU767	NSN555667	Hana Patel	hmutothere@geignone.com	NZU767	S67
WPC46	Warp Plasma Containment	Design magnetic containment for warp plasma	01/05/2026	26/07/2026	\$12,500.00	WU1955	APL67	Gravity and Physics	NZU404	NSN24996	Liam Faulkner	liamian@fipso.co.nz	NZU404	CRO17
										NSN007007	Hemi Bond	hmapzy@gotcha.com	NZU767	WPC46
Student_Course					Professor_Course									
NSN(FK)	CourseID(FK)				ProfessorID(FK)	CourseID(FK)								
NSN051155	AE606				WU1955	AE606								
NSN051155	TM888				WU1955	TM888								
NSN897654	APL67				WU1643	APL67								
NSN211015	ME404				WU1905	ME404								
NSN24996	ME404				WU1701	WT808								
NSN555667	WT808													
NSN007007	WT808													

APA References

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